

LUTSELK'E, NT, Canada

WMO: 715290

Lat: 62.416N

Lon: 110.690W

Elev: 179

StdP: 99.19

Time Zone: -7.00 (NAM)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.5	(c) -37.4	(d) -44.3	(e) 0.0	(f) -39.4	(g) -42.0	(h) 0.1	(i) -37.3	(j) 9.5	(k) -21.7	(l) 8.6	(m) -20.4	(n) 2.1	(o) 70	(p) 0.765

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	25.5	15.8	23.5	14.9	21.7	14.2	17.0	23.2	15.9	21.8	14.9	20.2	3.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	10.5	18.8	13.3	9.7	18.2	12.2	9.1	17.5	48.4	23.2	45.3	21.8	42.5	20.3	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.3	(b) 8.1	(c) 7.1	(e) -41.1	(f) 28.7	(g) 2.3	(h) 1.8	(i) -42.7	(j) 30.0	(k) -44.1	(l) 31.1	(m) -45.4	(n) 32.1	(o) -47.0	(p) 33.4		
(a) 9.3	(b) 8.1	(c) 7.1	(e) -41.2	(f) 19.3	(g) 2.3	(h) 1.2	(i) -42.8	(j) 20.2	(k) -44.2	(l) 20.9	(m) -45.4	(n) 21.6	(o) -47.1	(p) 22.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.4	-24.4	-23.3	-17.1	-7.0	3.1	11.7	15.5	13.0	7.4	-0.6	-10.9	-21.0		
	DBStd	15.54	8.08	7.49	9.03	7.99	6.02	4.52	3.61	3.80	4.30	4.46	5.83	8.14		
	HDD10.0	5641	1066	933	839	511	224	33	2	13	101	330	627	961		
	HDD18.3	8309	1324	1166	1098	761	472	203	103	170	327	588	877	1219		
	CDD10.0	398	0	0	0	0	10	84	173	107	23	1	0	0		
	CDD18.3	24	0	0	0	0	0	4	15	6	0	0	0	0		
	CDH23.3	181	0	0	0	0	1	31	104	44	1	0	0	0		
	CDH26.7	22	0	0	0	0	0	2	14	6	0	0	0	0		
	WSAvg	3.2	2.6	2.7	3.1	3.6	3.9	3.2	2.9	3.0	3.4	3.4	3.4	2.7		
	PrecAvg	351	15	12	16	14	24	41	53	57	46	31	24	17		
(15) Precipitation	PrecMax	458	29	23	31	28	46	95	106	101	91	59	39	27		
	PrecMin	255	2	3	7	3	4	8	20	23	15	10	11	8		
	PrecStd	38	7	4	6	5	11	21	18	17	18	10	7	5		
	PrecStd	38	7	4	6	5	11	21	18	17	18	10	7	5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.3	-2.4	6.2	12.6	21.9	26.4	28.7	27.2	21.1	13.1	3.3	-2.1		
		MCWB	-3.4	-3.4	2.2	6.2	12.2	15.7	17.0	17.2	14.4	8.7	1.6	-2.9		
	2%	DB	-5.6	-7.0	2.2	9.1	18.0	23.9	26.0	24.3	18.3	9.4	0.7	-4.5		
		MCWB	-6.1	-7.6	-0.3	4.2	9.5	14.2	16.2	15.7	12.4	6.1	-0.3	-5.3		
	5%	DB	-9.5	-9.7	-0.5	6.5	14.9	21.9	24.1	22.0	16.0	7.1	-1.0	-7.0		
		MCWB	-9.9	-10.3	-2.2	2.7	8.0	13.4	15.5	14.7	11.2	5.0	-1.8	-7.7		
	10%	DB	-12.5	-12.2	-3.5	4.0	12.1	19.7	22.2	19.9	14.0	5.2	-2.9	-9.3		
		MCWB	-12.9	-12.7	-4.9	1.3	6.4	12.3	14.8	14.0	10.5	3.5	-3.6	-9.9		
	0.4%	WB	-3.3	-3.4	2.4	6.6	12.4	16.9	18.6	18.0	14.8	9.2	1.6	-2.9		
		MCDB	-2.4	-2.4	5.4	11.8	20.5	23.4	25.3	25.5	19.8	12.8	2.9	-2.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-6.1	-7.6	-0.2	4.4	9.9	15.2	17.4	16.6	13.0	6.4	-0.2	-5.2		
		MCDB	-5.6	-7.0	2.0	8.7	17.3	21.8	23.7	22.4	17.2	8.5	0.7	-4.5		
	5%	WB	-9.9	-10.3	-2.2	2.7	8.3	14.0	16.4	15.5	11.8	5.1	-1.8	-7.6		
		MCDB	-9.6	-9.7	-0.5	6.1	14.6	20.5	22.5	20.4	15.4	7.3	-1.0	-7.0		
	10%	WB	-12.9	-12.7	-4.8	1.4	6.7	12.9	15.4	14.5	10.6	3.7	-3.6	-9.9		
		MCDB	-12.5	-12.2	-3.5	3.8	11.6	19.2	21.1	18.9	13.6	5.1	-2.8	-9.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	9.1	11.1	11.6	10.2	10.7	10.0	9.0	6.9	5.0	5.3	7.2		
		MCDBR	8.7	10.2	11.9	12.4	14.4	13.4	12.4	12.5	10.3	7.7	5.4	7.2		
		MCWBR	8.4	9.8	9.9	8.5	8.4	7.2	6.4	6.5	6.1	5.6	5.1	6.9		
		MCDBR	8.8	10.2	11.6	11.6	13.5	12.4	10.9	10.8	9.5	7.3	5.4	7.2		
	5% WB	MCWBR	8.4	9.8	9.8	8.2	8.1	7.1	6.1	6.1	6.0	5.5	5.1	6.9		
		MCWBR	8.4	9.8	9.8	8.2	8.1	7.1	6.1	6.1	6.0	5.5	5.1	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.181	0.235	0.258	0.307	0.337	0.349	0.371	0.343	0.308	0.266	0.191	0.147		
	taud		2.052	2.241	2.274	2.251	2.413	2.439	2.425	2.489	2.576	2.437	2.046	1.946		
	Ebn at Noon		539	754	866	880	882	875	842	840	814	713	495	384		
	Edh at Noon		32	59	84	107	101	100	100	86	65	49	31	18		
(44) All-Sky Solar Radiation	RadAvg		0.28	1.07	2.79	4.91	5.71	6.05	5.40	4.01	2.42	0.93	0.28	0.12		
	RadStd		0.03	0.09	0.20	0.27	0.41	0.47	0.23	0.24	0.19	0.10	0.03	0.01		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page